

THE PLAIN-ENGLISH COMMERCIAL ROADMAP

Building an Unbeatable Silicon and Software Stack for Physical AI (2026–2029)

1. Executive Summary: The Landscape of Physical AI Silicon

Artificial intelligence is moving away from purely digital cloud systems to physical devices deployed at the edge—like humanoids, mobile robots, and smart home appliances. While giant cloud companies are spending heavily on servers, mobile devices have strict limits on battery, heat, and space. Standard digital computer chips drain batteries too quickly and run too hot when trying to make real-time decisions. This has created a massive market opportunity for ****Analog In-Memory Computing (IMC)****, which runs calculations using continuous electrical currents inside memory blocks instead of shuttling massive amounts of digital data back and forth. This document lays out our 36-month strategic execution roadmap to successfully build and sell this chip to the physical AI market.

The Core Venture Opportunity: While standard digital chips scale well, they waste up to 90% of their energy moving data between processing units and memory blocks. Our analog chip eliminates this waste by running calculations directly where the data lives, cutting power use down to sub-watt levels. This allows robots to execute complex visual and movement actions locally on a tiny battery.

2. The New Paradigm: Lightweight, Adaptive Robot Brains

Modern robots no longer run on fixed, pre-programmed rules. Instead, they use lightweight visual-motor models (known as Vision-Language-Action, or VLA models) that need to adapt in the field. Models like Google's On-Device 2 and Microsoft's Rho-alpha can learn entirely new physical robot designs, hand shapes, or environments in just a few hours using under 200 demonstration examples. For our analog startup, this means our silicon cannot be designed around fixed digital calculations. We must support dynamic, frequent weight updates on-chip and have a software-smart compiler that translates neural network layers effortlessly.

3. The 36-Month Strategic Roadmap

To build commercial value, win venture funding, and deliver a reliable chip, we will execute our roadmap in three carefully structured phases over the next three years:

Phase 1 (Months 0–9): Software-Aware Co-Design & Training

Before we spend millions manufacturing physical silicon, we must build a perfect virtual replica of our analog chip in software. We will write a PyTorch and JAX-compatible simulator that models all of our chip's physical non-idealities (such as electrical noise, voltage decay, and heat-induced drift). This allows robot developers to train their models using Hardware-Aware Training (HAT). Because the AI learns to expect these physical variations during training, it remains highly accurate when loaded onto our physical, imperfect hardware.

Phase 2 (Months 9–18): Building a Hybrid, Multi-Part Chip

We will avoid building a 'purely' analog chip, as analog is terrible at making complex, dynamic decisions. Instead, we will build a **hybrid chiplet**. We will use our low-power analog arrays to run heavy, static mathematical calculations, but place a small, high-performance digital coprocessor right next to them on the silicon to handle dynamic decisions and routing. To allow fast on-device learning in the field, we will transition away from slow Flash memory to high-endurance, fast-writing magnetic memory (STT-MRAM or FeRAM) that can be reprogrammed billions of times at standard operating voltages.

Phase 3 (Months 18–36): Real-World Integration & Production Scaling

We will integrate our virtual chip simulator directly with physically accurate digital twins, like NVIDIA Isaac Sim. This lets robot developers train, verify, and safety-test their AI inside a high-fidelity virtual environment before deploying it on physical machines. We will perform Hardware-in-the-Loop (HIL) testing, linking our physical chip prototypes to deterministic safety controllers (FPGAs) to prove our millisecond latency loops are safe. Finally, we will partner with top-tier semiconductor fabs to scale our RRAM/memristor designs from small research lines to standard 300mm production wafers, establishing high-volume manufacturing.

4. Solving the Six Physical and Engineering Challenges

To commercialize our analog silicon, our hardware and compiler teams must solve six specific physical, thermal, and micro-architectural challenges. We have mapped out a plain-English playbook for each challenge, which is fully integrated as core milestones in our 36-month roadmap.

Challenge 1: Stop Digital Noise from Leaking into Analog Math

The Problem (Substrate Noise Coupling): When high-speed digital circuits on the chip switch off and on, they inject high-frequency electrical noise into the shared silicon base. This noise leaks laterally into our sensitive analog arrays, changing the voltages and ruining the mathematical accuracy of our AI calculation.

Our Solution: We will implement *Triple-Well Isolation*. This acts like a deep protective moat, electronically separating the digital and analog sections of our silicon to block lateral noise. For our long-term product lines, we will migrate to a *glass package*. Glass acts as an excellent insulator, completely eliminating lateral electrical noise between different sections of the chip.

Challenge 2: Heat Management & Physical Drift

The Problem (Electro-Thermal Drift): Running the chip generates heat. As the temperature rises, the electrical resistance of our analog memory materials changes, causing the programmed weight values to drift and degrade accuracy. Additionally, placing thin insulating layers on the chip traps heat, raising local channel temperatures.

Our Solution: We will run fast thermal calculations during early layout design to avoid placing hot digital parts near quiet analog arrays. We will also integrate *leaf-vein cooling channels* directly into the package, using a split fluid network that cools down hot sections adaptively, lowering temperatures by up to 76% under maximum workloads.

Challenge 3: Training the AI to Survive Hardware Decay

The Problem (Material Aging & Conductance Drift): Over months of real-world use, analog memory materials naturally age and decay. This gradual drift in electrical properties degrades the AI's accuracy.

Our Solution: We will run *Hardware-Aware Training (HAT)* in software. By injecting simulated programming variations and material decay directly into the training loop, the optimizer is forced to search for broad, flat regions in the math where small changes in conductance do not affect the output. This ensures our chip maintains high accuracy over years of deployment without needing frequent, slow recalibration.

Challenge 4: Share the Signal Translators in Parallel

The Problem (ADC Group Parallelism): While analog math is nearly instantaneous, converting the output currents back to digital signals requires Analog-to-Digital Converters (ADCs). ADCs are bulky and expensive, so multiple memory columns must share a single ADC. If they have to wait in line to translate their signals, the chip's throughput stalls.

Our Solution: We will implement a hardware-compiler co-design called *Group-Level Parallelism (GLP)*. Our compiler will analyze the neural network and group independent operations so they run on different column groups. This allows all ADCs across the array to run in parallel, boosting overall chip speed by up to 5.74×.

Challenge 5: Fast Filtering to Prevent Power Waste

The Problem (Perceptual Overload): Robots process massive streams of visual and tactile data. Converting all of this data through power-hungry ADCs just to run self-attention layers drains the battery rapidly.

Our Solution: We will place a low-power, 4-bit analog filtering block directly on-chip before the ADCs. This analog filter evaluates incoming signals and immediately throws away 75% of useless, unchanging background data without ever invoking the power-hungry ADCs. Only the remaining 25% of important, changing data is converted to digital for exact processing, saving massive amounts of power.

Challenge 6: Build a Zero-Touch Translator

The Problem (The Compiler Wall): Robotics programmers write code in standard digital frameworks like JAX and PyTorch. They do not know how analog circuits work, and they will refuse to use our chip if they have to manually program electrical voltages.

Our Solution: We will build a customized MLIR-based compiler (analog-mlir). This software acts as an automatic translator: it takes standard AI code, automatically slices it, routes heavy math to our analog arrays, routes decision logic to our digital coprocessors, and configures the chip registers automatically behind the scenes.

5. Tactical Silicon Comparison Matrix

This matrix outlines where our Analog IMC approach holds an unassailable physical advantage versus incumbent edge silicon across key physical AI parameters:

Silicon Class	Typical Power	Latency / Reflexes	Primary Limitations	Best Fit
Central SoCs (NVIDIA Jetson)	30W - 100W	10ms - 50ms (Slow)	High cost, heavy, requires active fan cooling	High-level reasoning, spatial world models
Reflex Controllers (Lattice FPGA)	Sub-1W	Sub-500ns (Instant)	Strict logic only, cannot run massive VLAs	Deterministic motor loops, functional safety
Analog IMC (Our Solution)	Sub-1W	Sub-1ms (Fast)	Weight writing is slow, requires hybrid digital coprocessor	Low-power VLA inference, local tactile/DVS fusion